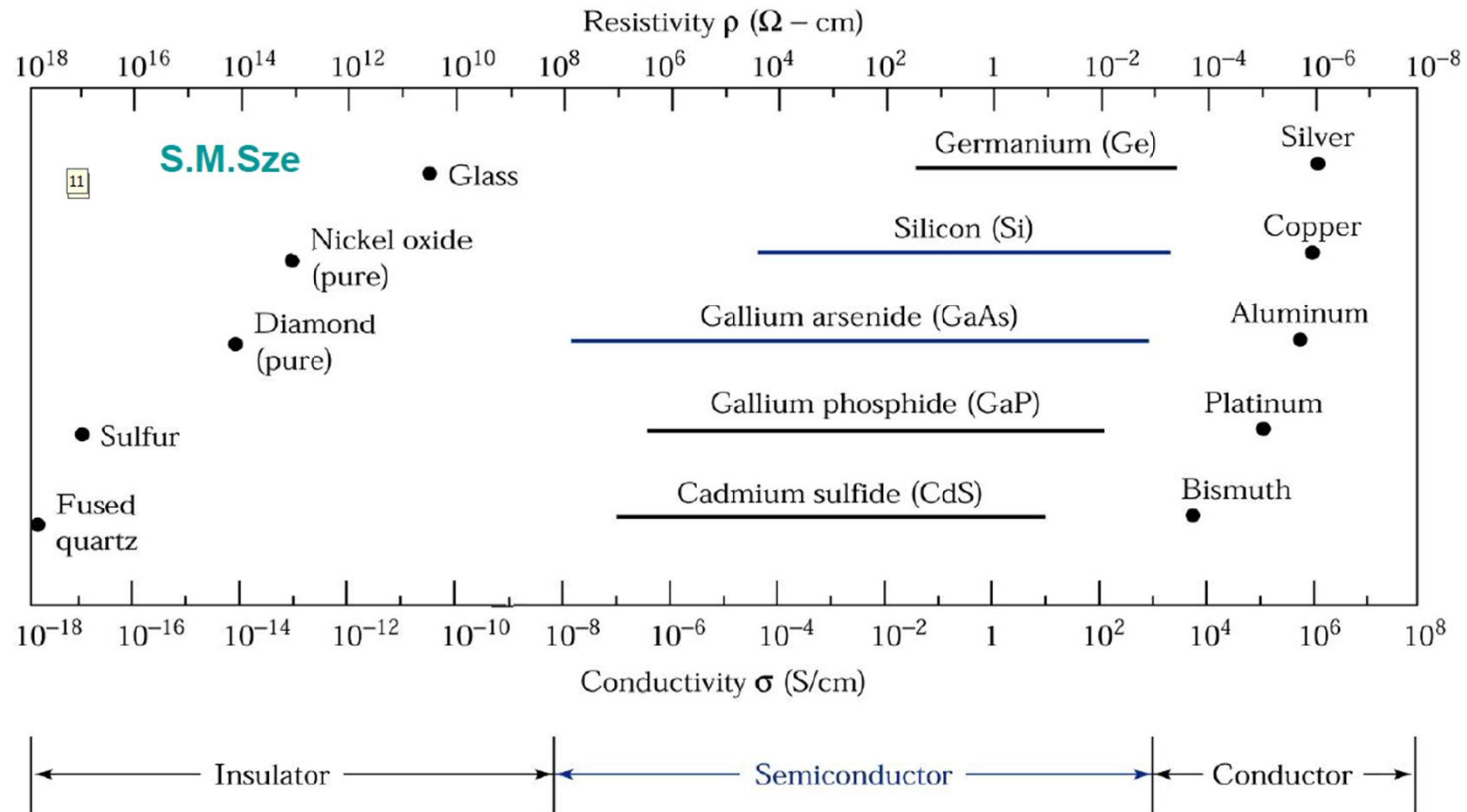


L1: Introduction to semiconductor devices

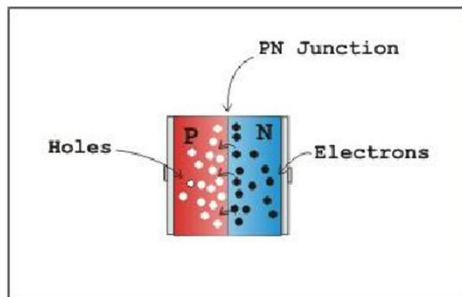


Dr. Shuming Chen

Department of Electrical and Electronic Engineering, SUSTech

Devices in Action

Basic devices



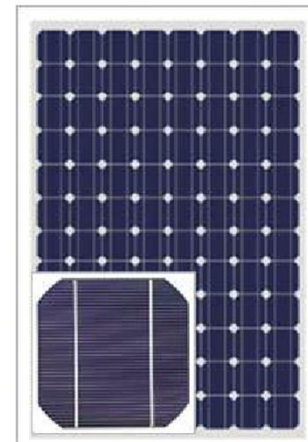
OLED Display



Rectifier diode



LED



Solar cell



Laser diode

Basic devices (two terminals)

Gunn Diode



LED

PIN Diode



Step Recovery Diode



Laser Diode



Photo Diode



Schottky Diode



Zener Diode

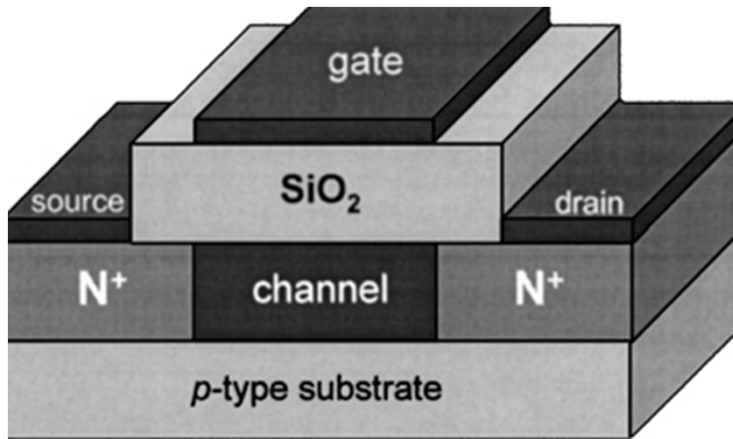
Tunnel Diode



Varactor Diode

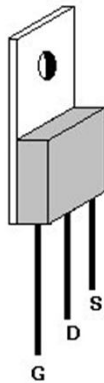


Basic devices (three terminals)



2N7000
TFET

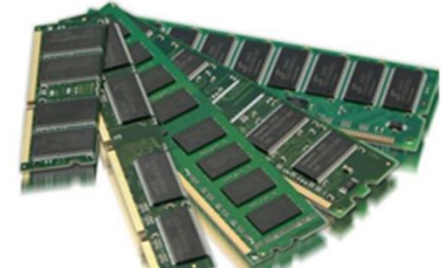
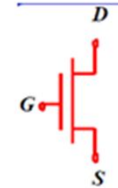
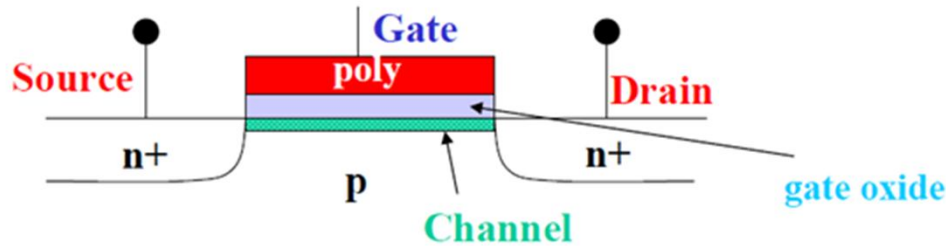
N-Channel
200 mAMPS
60 VOLTS
 $R_{DS(on)} = 5 \text{ ohm}$



IRFZ44N
POWER MOSFET
N-Channel
50 Amp
60 VOLTS
 $R_{DS(on)} = 0.028 \text{ ohm}$



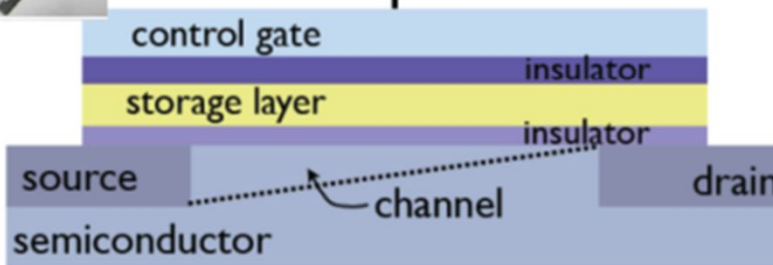
Basic devices (three terminals)



the basics of Flash

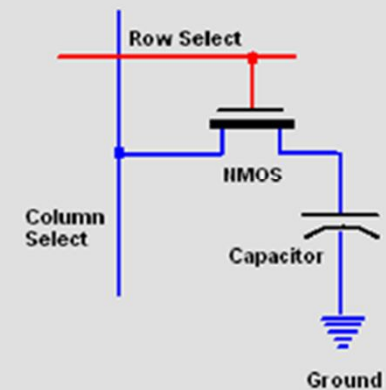


positive V - channel pinched
negative V - channel open



- electrons go between source and drain
- 'control gate' regulates how many negative voltage stops electrons, positive helps

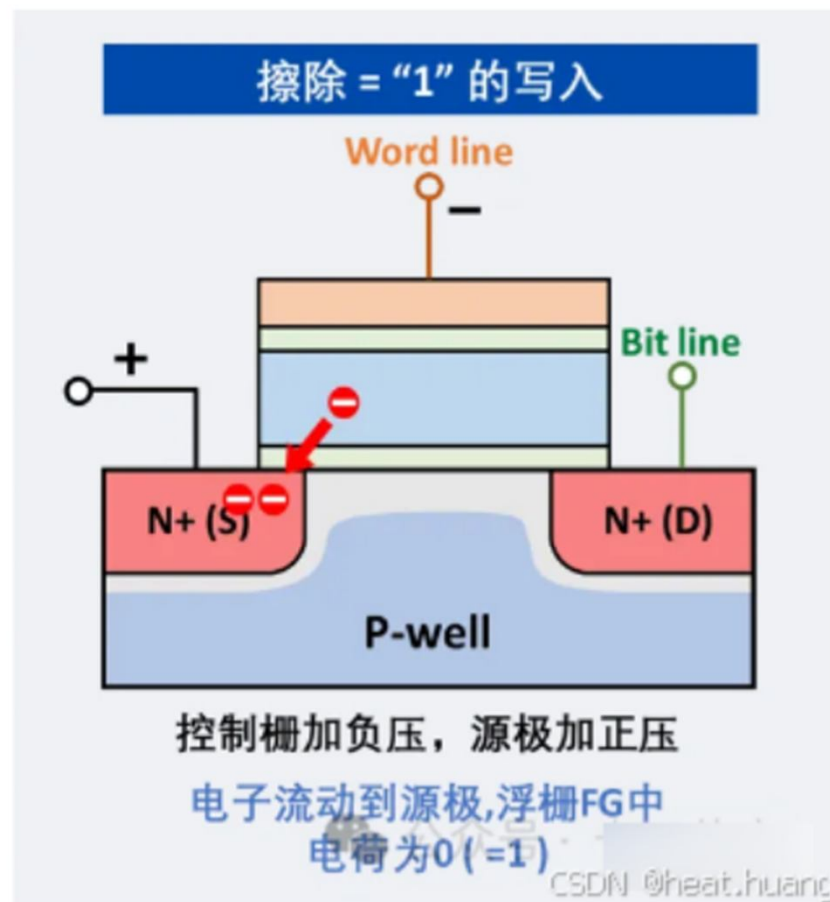
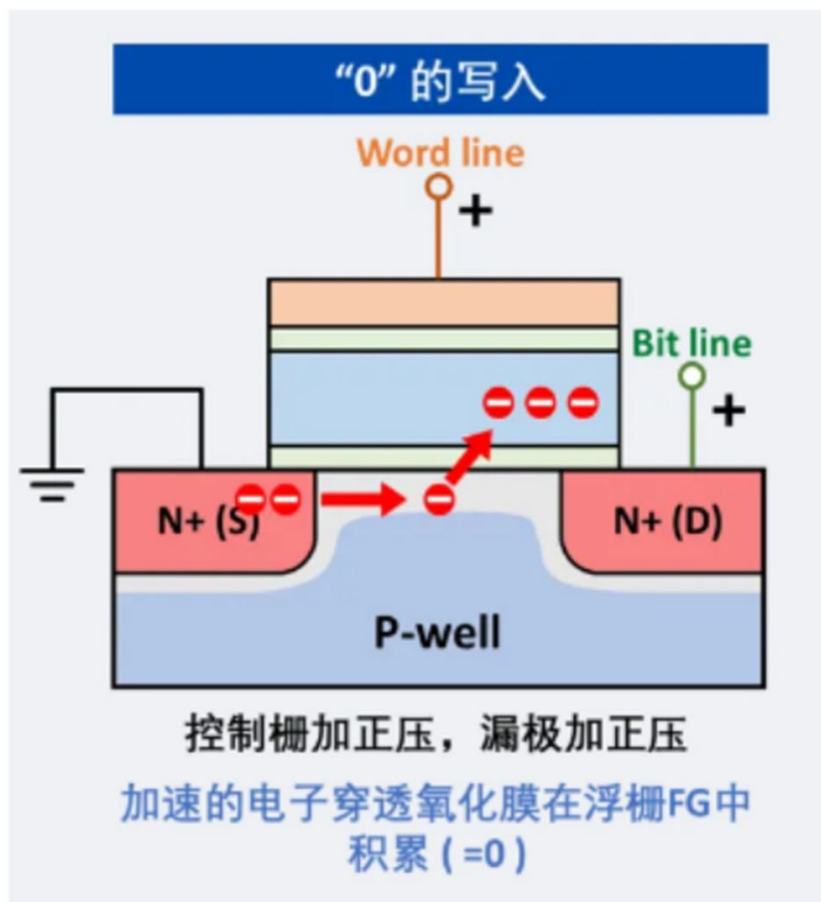
One Dynamic RAM Bit (DRAM bit)



Basic devices (Flash)

- 写入和消除工作原理

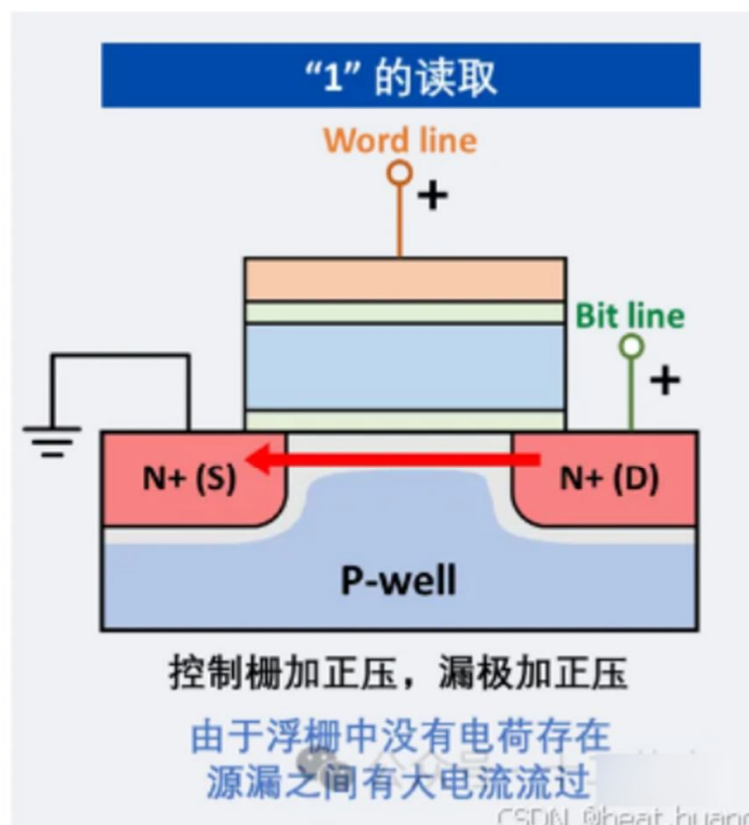
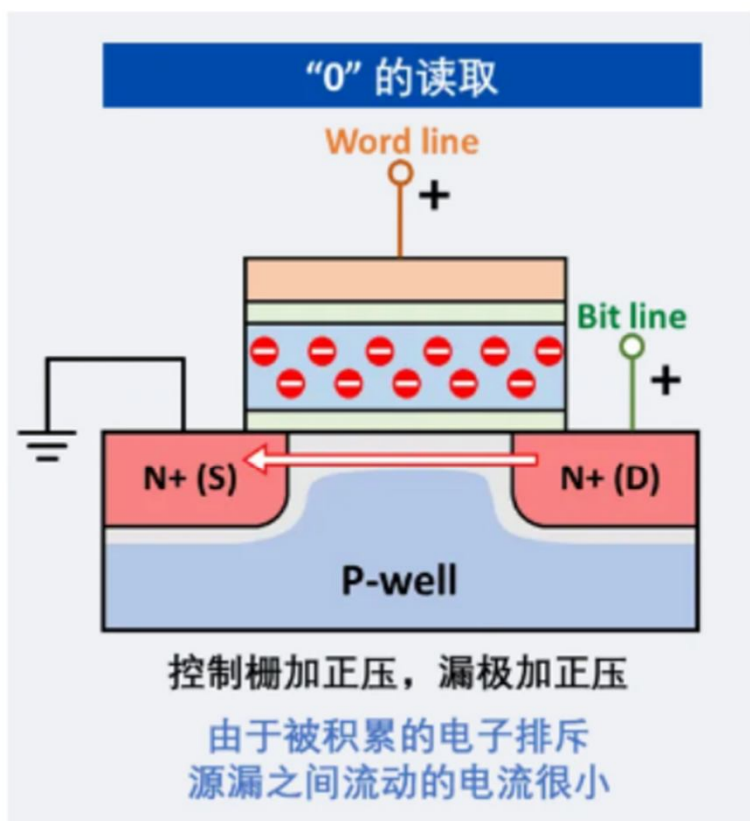
闪存中写入“0”相当于在浮置栅极中积累电荷。写入“1”是抽出浮置栅极电荷的操作，相当于“擦除”。



Basic devices (Flash)

- 读取工作原理

闪存的读取是在控制栅极施加正电压，检测源极和漏极之间电流的大小来判断是“0”还是“1”。当浮置栅极积累有电荷（“0”）时，电流难以流动；当没有电荷（“1”）时，会有较大的电流流过。

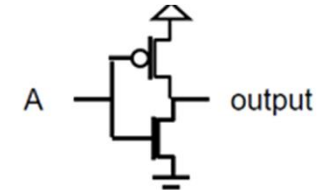


From devices to logic gates

Inverter

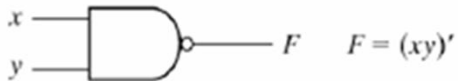


x	F
0	1
1	0

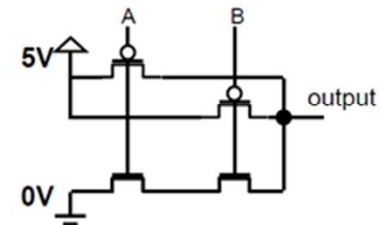


NOT Gate (Inverter)

NAND



x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

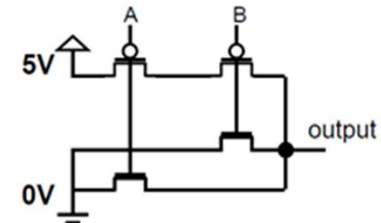


NAND Gate

NOR



x	y	F
0	0	1
0	1	0
1	0	0
1	1	0



NOR Gate

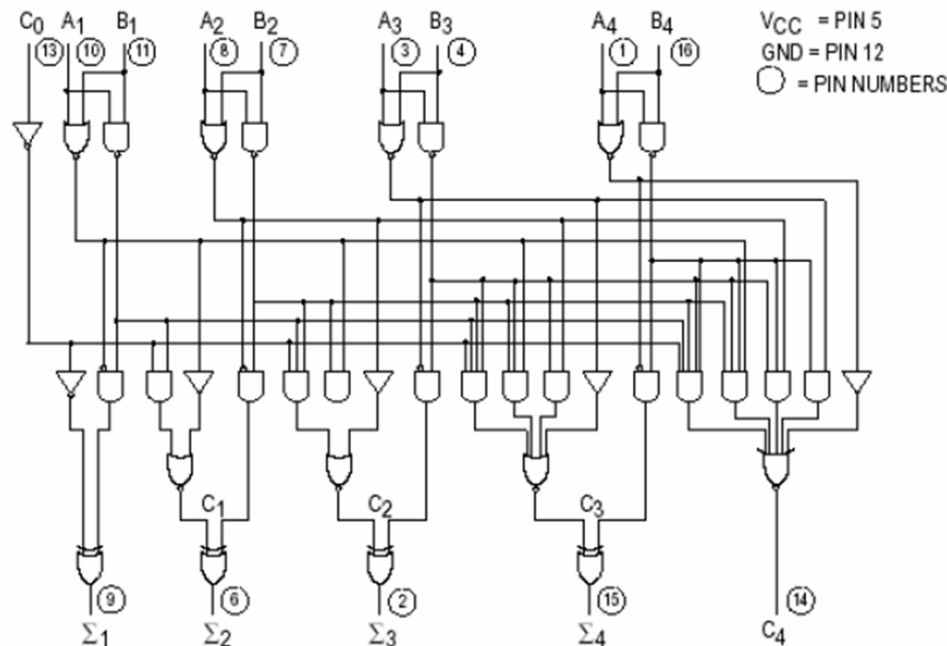


From logic gates to integrated circuit

TTL Example: 4-bit Binary Adder

4-BIT BINARY FULL ADDER WITH FAST CARRY

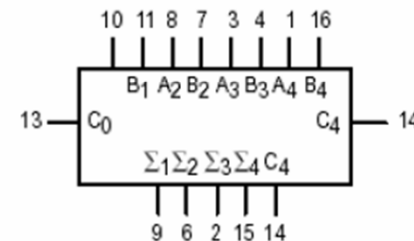
The SN54/74LS83A is a high-speed 4-Bit binary Full Adder with internal carry lookahead. It accepts two 4-bit binary words (A_1 – A_4 , B_1 – B_4) and a Carry Input (C_0). It generates the binary Sum outputs Σ_1 – Σ_4 and the Carry Output (C_4) from the most significant bit. The LS83A operates with either active HIGH or active LOW operands (positive or negative logic). The SN54/74LS283 is recommended for new designs since it is identical in function with this device and features standard corner power pins.



SN54/74LS83A

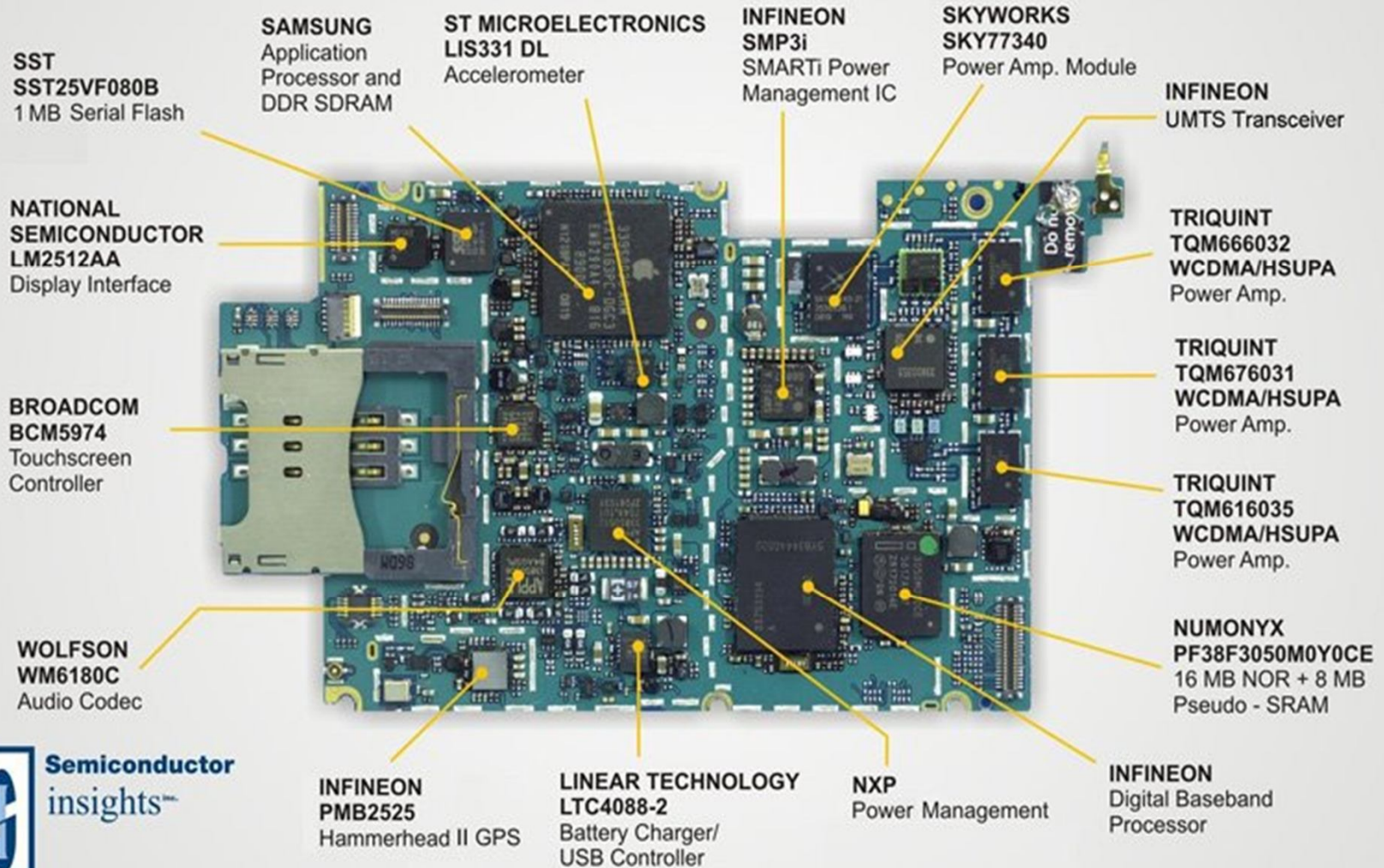
4-BIT BINARY FULL ADDER
WITH FAST CARRY
LOW POWER SCHOTTKY

LOGIC SYMBOL



From integrated circuits to real products

IPHONE 5S mainboard



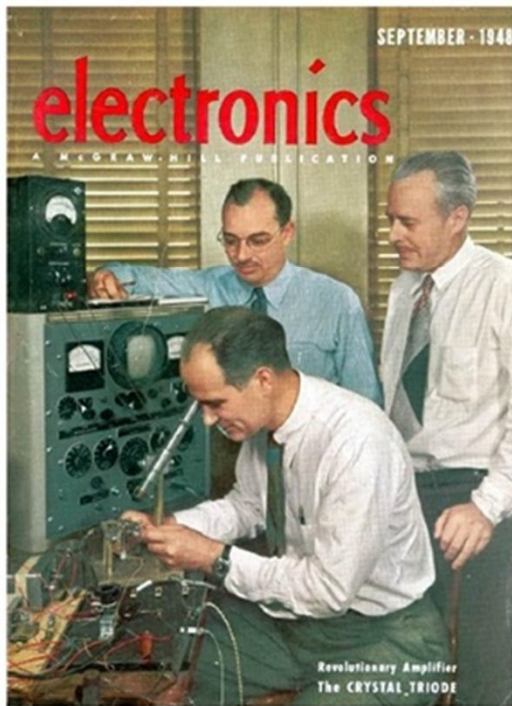
History and Moore's law

How old is this field ?

1st Transistor

Christmas Eve 1947

By Bardeen, Brattain, and Shockley,
Nobel Laureates in Physics 1956



威廉·肖克利和同事在贝尔实验室共同发明了晶体管。不过肖克利的心中有一个富翁梦，在贝尔实验室又无法实现这个梦想，于是他决定回到老家圣克拉拉谷（即后来的硅谷）创办自己的公司。

(http://www.bellsystemmemorial.com/belllabs_transistor.html)

美国贝尔实验室的肖克利、巴丁和布拉顿

The Traitorous 8 (仙童8叛逆) and Birth of Silicon Valley

The **traitorous eight** was a group of eight employees who left [Shockley Semiconductor Laboratory](#) in 1957 to found [Fairchild Semiconductor](#).

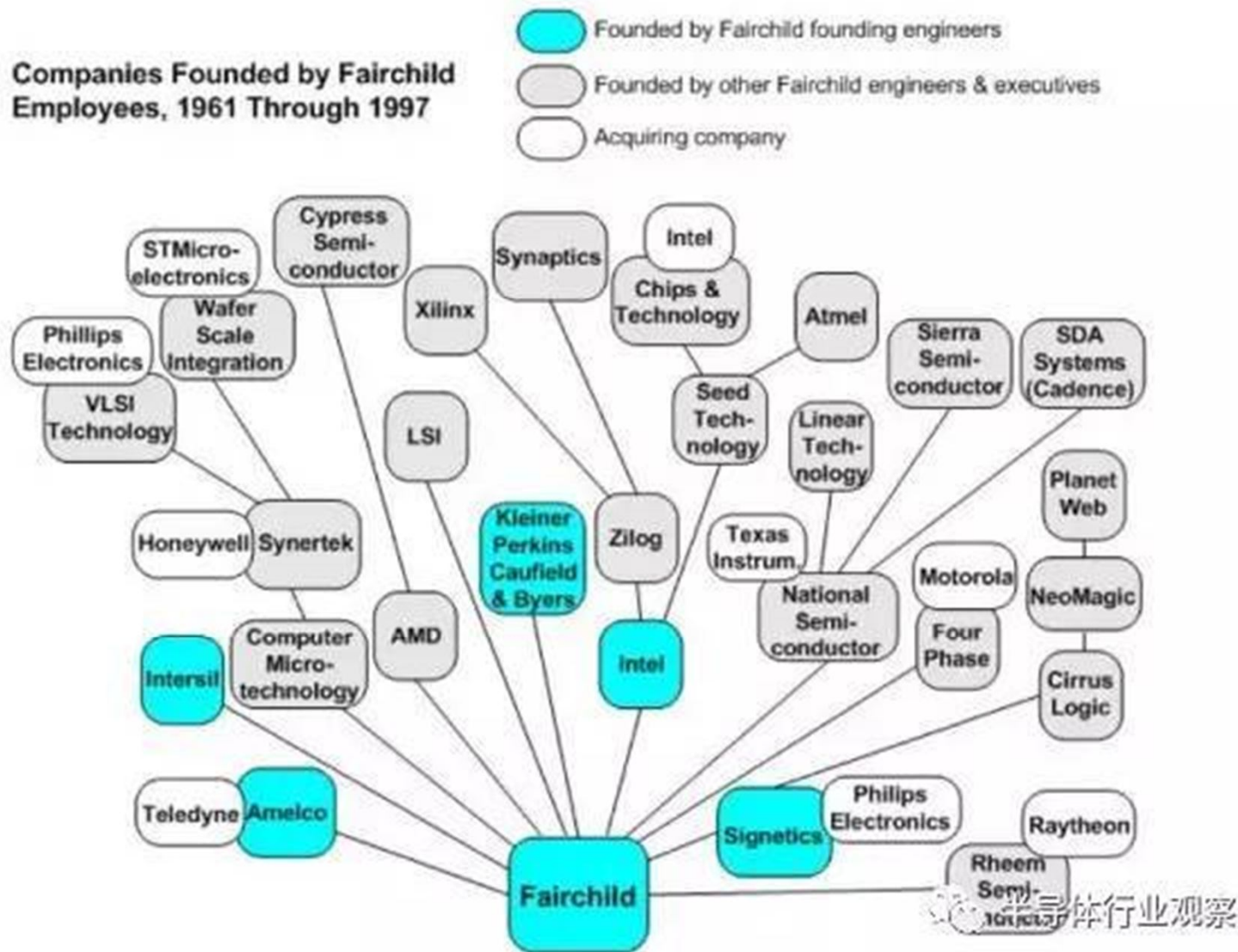
1968, Moore and Noyce founded Intel

Jerry Sanders, along with seven of his colleagues from Fairchild, on May 1, 1969 founded AMD



From left to right: Gordon Moore, C. Sheldon Roberts, Eugene Kleiner, Robert Noyce, Victor Grinich, Julius Blank, Jean Hoerni and Jay Last (1960)

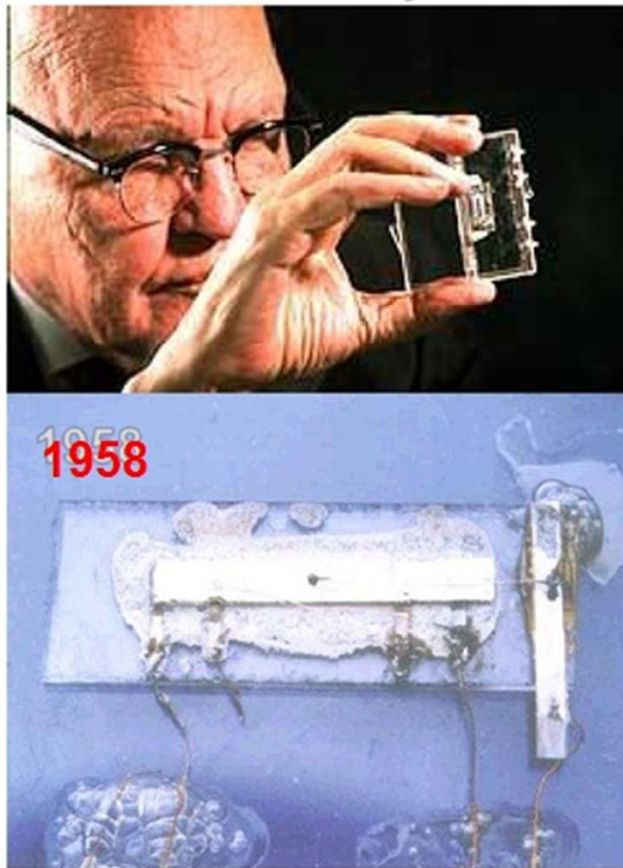
The Traitorous 8 (仙童8叛逆) and Birth of Silicon Valley



The first IC

1st Integrated Circuit

By Kilby,
Nobel Laureate in Physics 2000
and
Robert Noyce



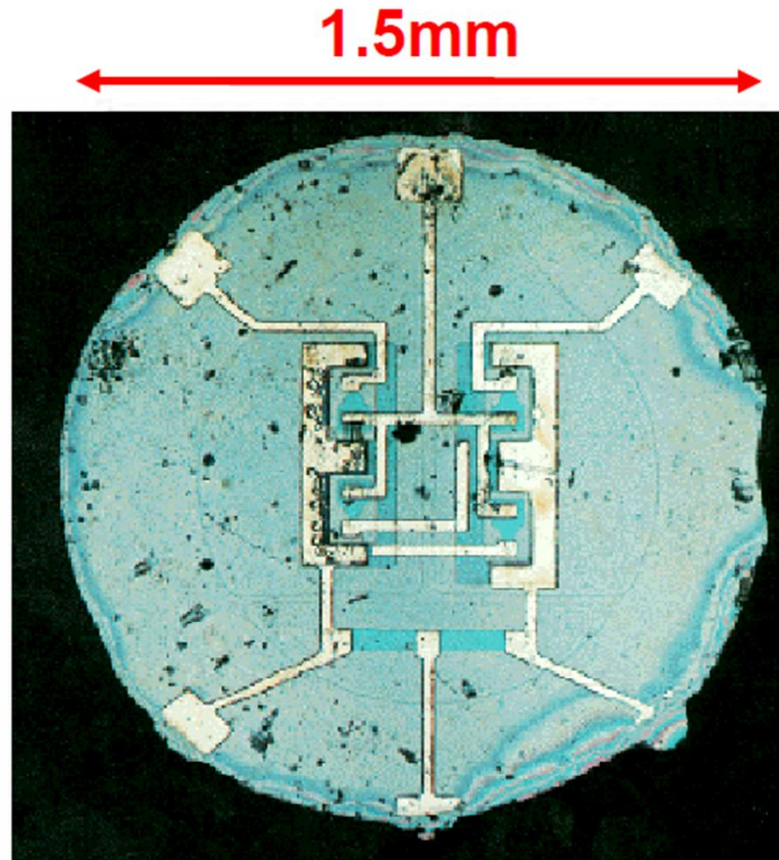
(Courtesy of TI and Huff, SEMATECH)

1958年，杰克·基尔比和罗伯特·诺伊斯这两个人几乎同时独立发明了集成电路。他们将晶体管、电阻和电容等集成在微小的平板上，用热焊方式把元件以极细的导线互连，在不超过4平方毫米的面积上，大约集成了20余个元件。1959年2月6日，基尔比向美国专利局申报专利，这种由半导体元件构成的微型固体组合件，从此被命名为“集成电路”（IC）。基尔比被誉为“第一块集成电路的发明家”，而诺伊斯被誉为“提出了适合于工业生产的集成电路理论”的人。1969年，美国联邦法院最后从法律上承认了集成电路是一项“同时的发明”。

How old is this field ?

First Silicon planar IC chip made by R. Noyce of Fairchild Camera in 1961

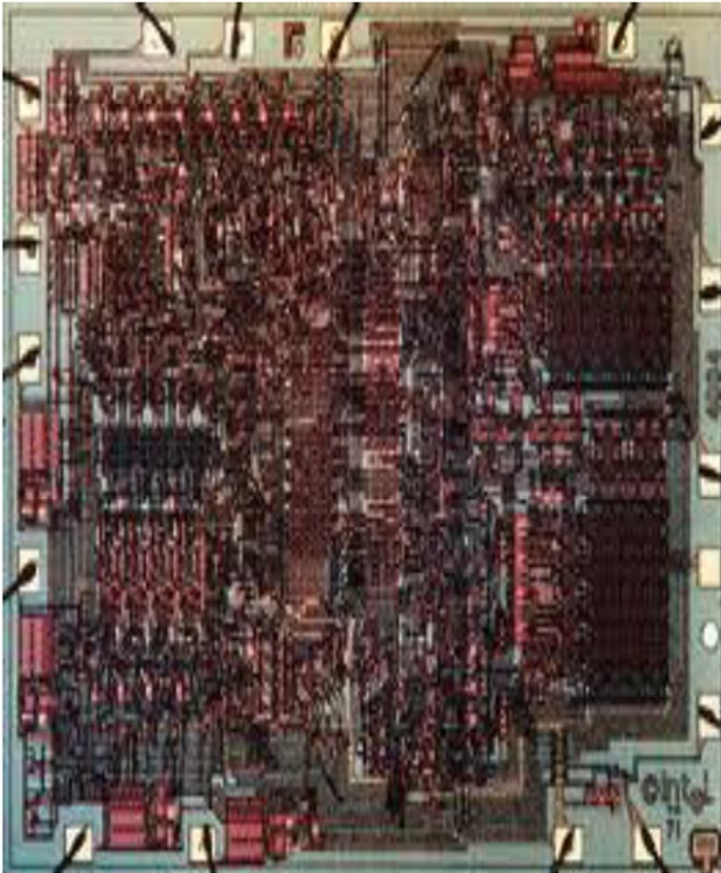
Photo courtesy:
Fairchild
Semiconductor
International



A flip-flop circuit containing 6 devices, produced in planar technology.

How old is this field ?

World 's first microprocessor by Intel



1971

A 4-bit microprocessor
Intel 4004.

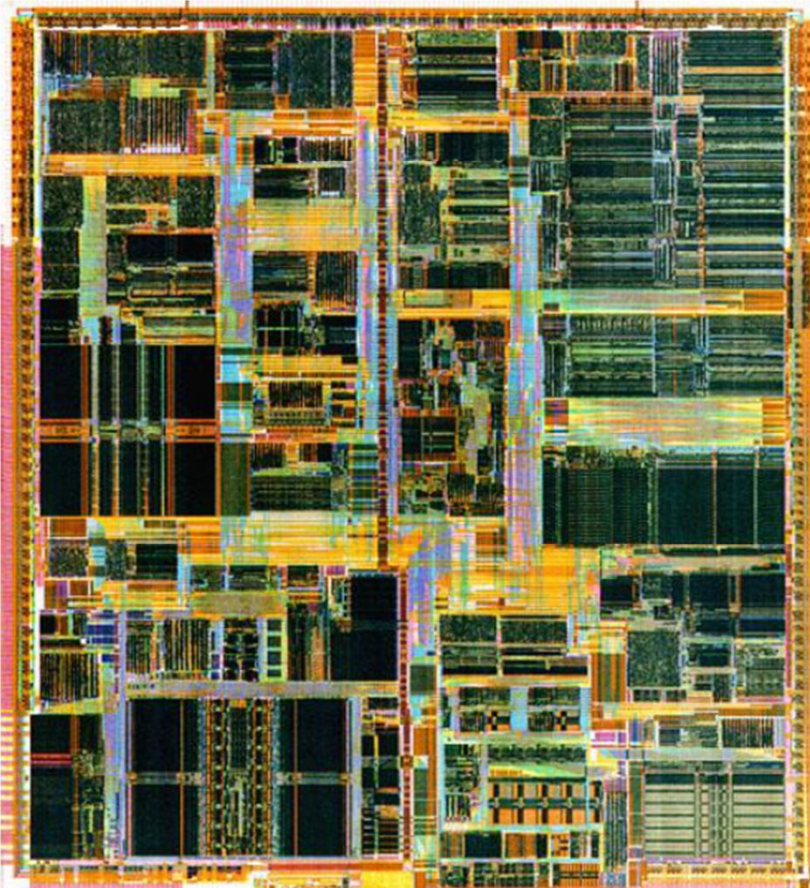
- 10 μm technology
- 3 mm $\times$ 4 mm
- 2300 MOS-FETs
- 108 kHz clock frequency

Source:

Intel Corporation

How old is this field ?

Pentium IV processor



2001

A ULSI chip with 32-bit processor *Intel Pentium 4*.

- 0.18 μ m CMOS technology
- 17.5 mm $\times$ 19 mm
- 42 000 000 components
- 1.6 GHz clock frequency

Source:

Intel Corporation

Moore's law

Moore's law is the observation that the number of [transistors](#) in a dense [integrated circuit](#) (IC) doubles about every two years. Moore's law is an [observation](#) and [projection](#) of a historical trend. Rather than a law of physics, it is an [empirical relationship](#) linked to gains from experience in production.



[Gordon Moore](#)
(1929-2023)

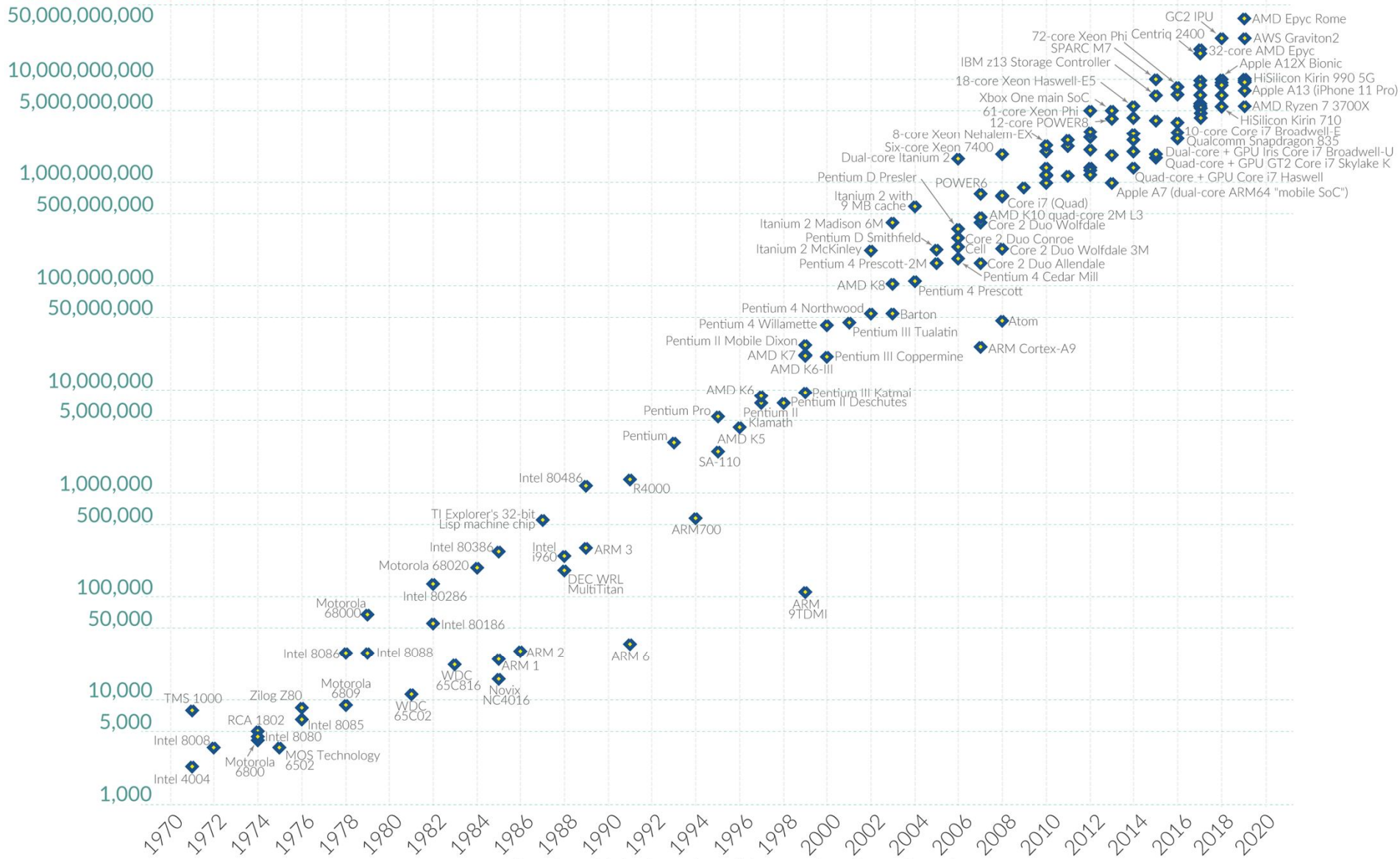
The observation is named after [Gordon Moore](#), the co-founder of [Fairchild Semiconductor](#) and CEO and co-founder of [Intel](#), who in 1965 posited a [doubling every year](#) in the number of components per integrated circuit, and projected this rate of growth would continue for at least another decade. In 1975, looking forward to the next decade, he revised the forecast to doubling every two years, a compound annual growth rate (CAGR) of 41%. While Moore did not use empirical evidence in forecasting that the historical trend would continue, his prediction held since 1975 and has since become known as a "law."

https://en.wikipedia.org/wiki/Moore's_law

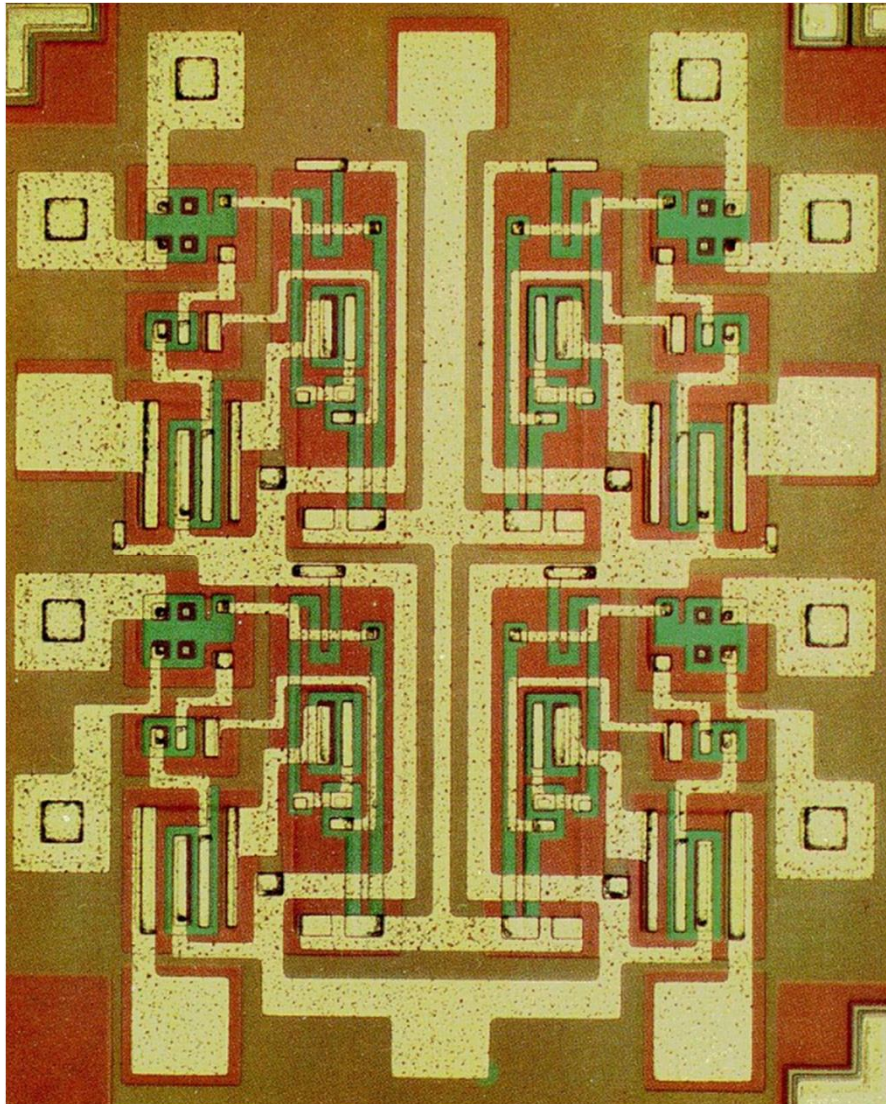
Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Transistor count



MOSFET scaling (process nodes)



MOSFET scaling (process nodes)

10 μm – 1971
6 μm – 1974
3 μm – 1977
1.5 μm – 1981
1 μm – 1984
800 nm – 1987
600 nm – 1990
350 nm – 1993
250 nm – 1996
180 nm – 1999
130 nm – 2001
90 nm – 2003
65 nm – 2005
45 nm – 2007
32 nm – 2009
22 nm – 2012
14 nm – 2014
10 nm – 2016
7 nm – 2018
5 nm – 2020
3 nm – 2022

SMIC中芯国际

TSMC台积电
(麒麟芯片)

It's a material world !

Semiconductor Devices and Materials

- Chips – Logic, Memory, Cellphones, ASICs, Sensors, Health Care, Displays, RFID tags, Entertainment
 - Silicon (in-production)
 - Germanium, III-V, Carbon (in-research)
- Wireless Circuits – Agile (Cellphones, Routers)
 - III-V (in production)
- LEDs
 - III-V (in-production)
- Solar Cells – Si, III-V, CIGS (in-production)

Electronics vs. Photonics

Electronics

- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si electronics is a **processing** based technology

Photonics

- GaAs, GaP, GaSb, InP, InAs, InSb, AlAs, GaN, AlN
- GaAs_xP_{1-x}, Al_xGa_{1-x}As, In_xGa_{1-x}P, In_xGa_{1-x}As, Al_xGa_{1-x}Sb, Al_xGa_{1-x}N, Ga_xIn_{1-x}N--**Ternary Alloys**
- In_xGa_{1-x}As_yP_{1-y}, Al_xIn_yGa_{1-x-y}As, In_xGa_{1-x}As_ySb_{1-y}, In_xGa_{1-x}N_yAs_{1-y}--**Quaternary Alloys**
- In_xGa_{1-x}N_yAs_{1-y-z}Sb_z--**Quinary Alloy**
- Heterojunctions are required for carrier confinement
- Photonics is a **materials** based technology